1 Knowledge-graph facts

PrimeKG
drug–disease edges

S — ciprofloxacin

R — contraindicated

O — myasthenia gravis

same facts, both relations

574 drugs · **1,722 edges**

indication vs. contraindication

matched within each drug, controlled for degree

2 Evidence search agent

Search agent

Wikipedia
574 articles

```
wiki.page(drug)
```

PubMed
40 abstracts / drug

```
pubmed.search(drug, k=40)
```

FDA label
openFDA, 472 drugs

```
openfda.label(drug)
```

name matcher

exact · redirect · MONDO ancestors

how plainly is the disease named?

0 · exact | 1 · alt. name | 2 · family | 3 · absent

chance floor: three unrelated diseases per drug give the rate a matcher reaches by luck (2.1–2.8%)

3 Models under test

Qwen2.5 0.5–14B | OLMo 2 7B | Granite 3.1 8B

closed book | **with the page**

Ciprofloxacin must not be given to patients who have …

A myasthenia gravis
B sibling disease
C sibling disease
D sibling disease
E cousin disease
F not enough information

*distractors are MONDO siblings; F lets the model **decline***

4 Within-drug analysis

Mantel–Haenszel OR

degree strata

gain from scale

effect of the page

Hetionet control

LABEL asymmetry: prose names contraindications **half as often** (11% vs 22%); the FDA label shows **none**.